AMONG US

AN AUTOMATED CLOSED-LOOP SORTING, SHOOTING & LIFTING TOY

MCT231 · BASICS OF MECHATRONICS

AMONG US

TEAM MEMBERS
Omar · Anas · Teefa · Essa · Mario

TARGET AGE
2 – 10 Years

THEME
Space / Reactor Defusal

TYPE
Activity-Based Toy

CONTROL
Dual MCU + Bluetooth

POWER
12V DC Single Source

PROJECT STORY

AMONG US is an interactive mechatronics toy where players stabilise a sabotaged lunar reactor by reproducing a colour sequence. The system automatically **sorts, shoots, and lifts** balls in a continuous closed-loop cycle while communicating with a Bluetooth application.

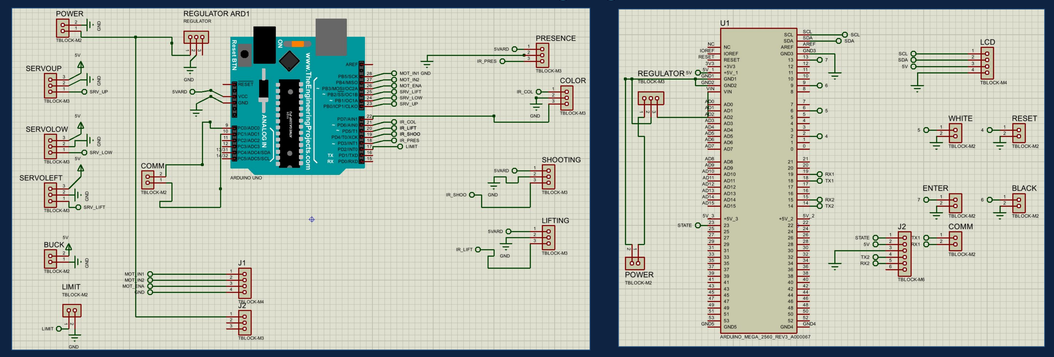
OBJECTIVES

- Colour-sequencing game
- Closed-loop automation (sort, shoot, lift)
- Dual interfaces: LCD + App
- Two microcontrollers communicating
- Safe, modular & easy to maintain

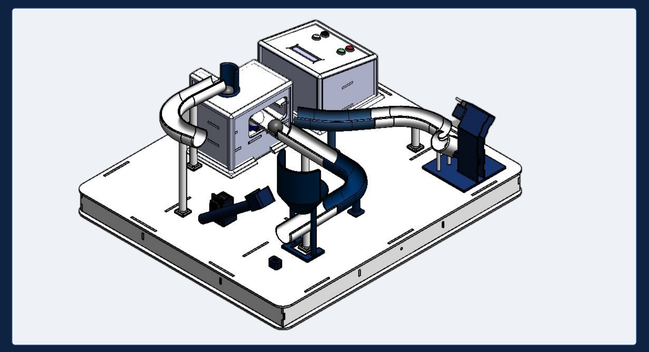
TARGET USERS

Children Aged 2 – 10
Enhances learning through play, patterns and cause & effect.

Wiring diagram



HERO IMAGE — FINAL PRODUCT (CAD RENDER)



SYSTEM ARCHITECTURE

SYSTEM BLOCK DIAGRAM

Bluetooth App (Processing)
↓
HC-05 Bluetooth
↓
ARDUINO MEGA (MASTER) UI · Logic · Comms — I2C 16×2 LCD; Buttons (W/B/Reset)
↓ UART · 9600
ARDUINO UNO (SLAVE) Sensors & Actuators
↓
IR Sensors ×5 · Servos ×3 L298N + DC Motor · Limit Switch

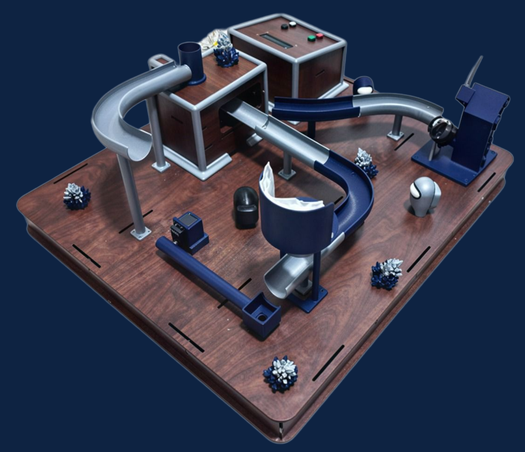
POWER ARCHITECTURE

220V AC → SMPS 12V DC → Buck → Servo Rail; +9V Reg → Uno VIN

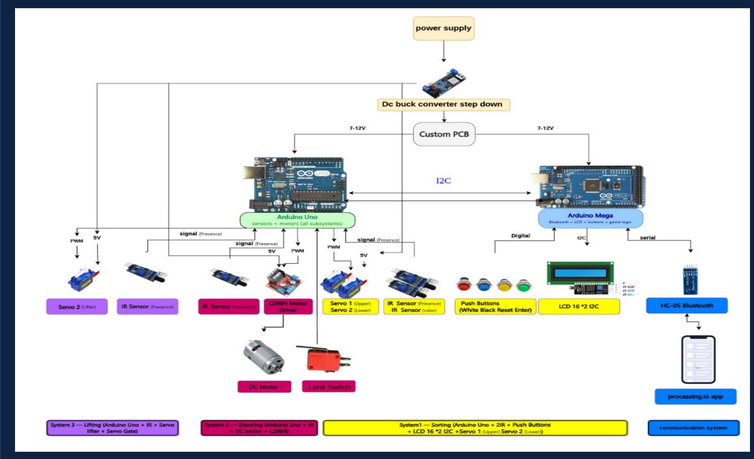
SYSTEM FLOWCHART

App Generates Target Sequence
↓
User Input (Buttons)
↓
Detect Ball (IR Sensors)
↓
Colour Match?
YES → Shoot to Funnel → Lift Back to Sorting Stage
NO → Return Path (Bypass) → Lift Back to Sorting Stage

FINAL PRODUCT

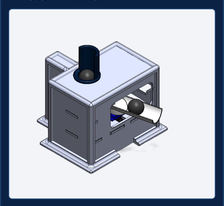


ELECTRICAL ARCHITECTURE



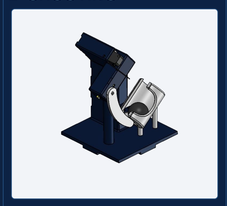
MECHANICAL SUBSYSTEMS

1 SORTING



- IR Color Sensor
- IR Presence Sensor
- Holder Servo
- Path Servo

2 SHOOTING



- IR Shooter Sensor
- DC Motor
- Firing Arm
- Limit Switch

3 LIFTING



- Lift Servo
- IR Lift Sensor

SOFTWARE & COMMUNICATION

COMMUNICATION PROTOCOL

FROM	TO	MSG	MEANING
App	Mega	W / B	Target sequence (White/Black)
App	Mega	R	Remote reset command
Mega	Uno	WBW…	Player sequence
Mega	Uno	R	Emergency reset
Mega	Uno	D	Done (all tasks complete)
Uno	Mega	C / W	Result (Correct / Wrong)
Mega	App	C / W	Result (Correct / Wrong)

APP INTERFACE (PROCESSING)



ENGINEERING CALC

Projectile Equation
v = √(R·g / sin 2θ)

Range R = 30 cm · Angle θ = 45°
g = 9.81 m/s²

Launch Velocity
v = 1.84 m/s

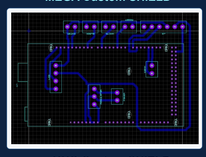
CORE SIZING CHAIN
profile → inertia → phase torques → RMS → SF 1.5

Servo ω_{max} = 4.189 rad/s
T_{RMS} = 7.05×10⁻³ N·m

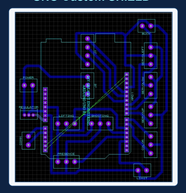
30 cm	Projectile Range
1.5	Safety Factor
< 200 ms	Actuator Response

CUSTOM PCB (SHIELD)

MEGA Custom SHIELD



UNO Custom SHIELD



VALIDATION RESULTS

- Colour sorting ≥ 95%
- Shooter range ≥ 30 cm
- Lift 100% return rate
- Closed-loop cycles ≥ 50
- Dual MCU communication
- Bluetooth App sync
- Custom PCB working
- Zero dropped UART msgs
- Safety measures passed

SAFETY FEATURES

Hidden Wiring · Emergency Reset · Low Voltage Design · Rounded Edges

DEMO VIDEO

GITHUB REPO

TECHNOLOGIES USED

Arduino · L298N Driver · HC-05 BT · I2C LCD
3D Printing · Laser-Cut MDF · Processing · Custom PCB

CONTACT

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